Supplementary Information to Closing the Loop: Integrating Material Needs of Energy Technologies into Energy System Models

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Appendix I
Scenarios

Fig. 1 shows the modelled links in different scenarios in comparison. Endogenous and exogenous aspects in the scenarios are summarized in table I.

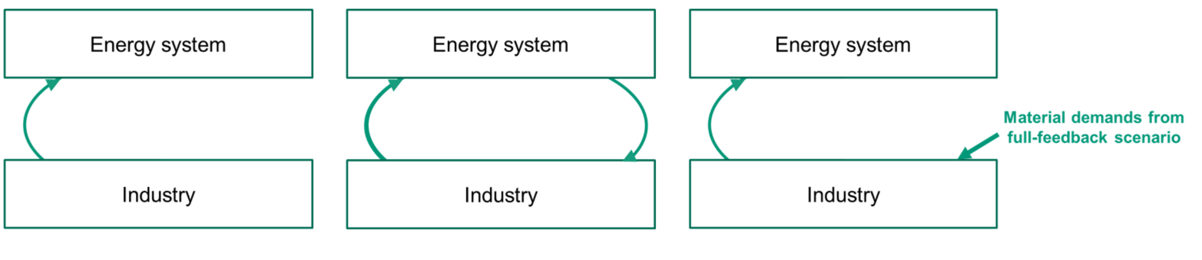

Fig. 1. Links included in the optimization of the scenarios: Reference (left), *100 % domestic* (middle) and *increased-material*s reference (right)

TABLE I

Endogenous and exogenous (input) parts of models in the compared scenarios.

Scenarios	**Endogenous**	**Exogenous (Inputs)**
all	Energy system and industry technology expansion and operation	Material demands not linked to technology expansion, Energy demands not linked to industry
reference (0 % domestic production)		additional: zero material demands for technology expansion
Domestic production (25 / 50 / 75 / 100 %)	Additional: material demands linked to technology expansion, according to domestic share	
Increased-material reference		Additional: material demands taken from 100% domestic scenario

APPENDIX II
MATERIAL FACTORS

Material factors for steel, cement and HVC are taken from [1] (electricity generators), [2] (process and space heat technologies), [3] (solar rooftop, storage technologies) and [4] (direct air capture). Material factors are shown in fig. 2 and fig. 3. Comparing the material factors for renewable generators, offshore wind generators show the highest bulk material demands per capacity, followed by onshore wind and solar. However, the actual maximum power output for these generators per capacity differ. When comparing the material demands per maximum average energy output of the technologies (i.e. running the technologies for one year at their maximum capacity factors), offshore wind AC remains the most material-intensive generator, followed by solar, offshore wind DC and onshore wind. For dispatchable generators, those equipped with carbon capture show substancially higher material factors.

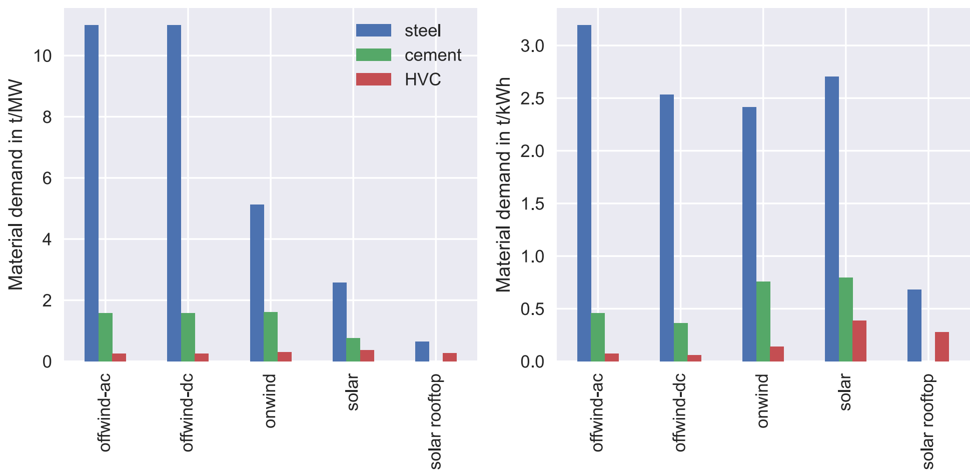

Fig. 2. Annualized material demand with equal distributed material demand over technology lifetime, relative to capacity (left) and maximum energy output (right). Maximum energy output means that the technologies are assumed to run at maximum capacity factor all year (no curtailment).

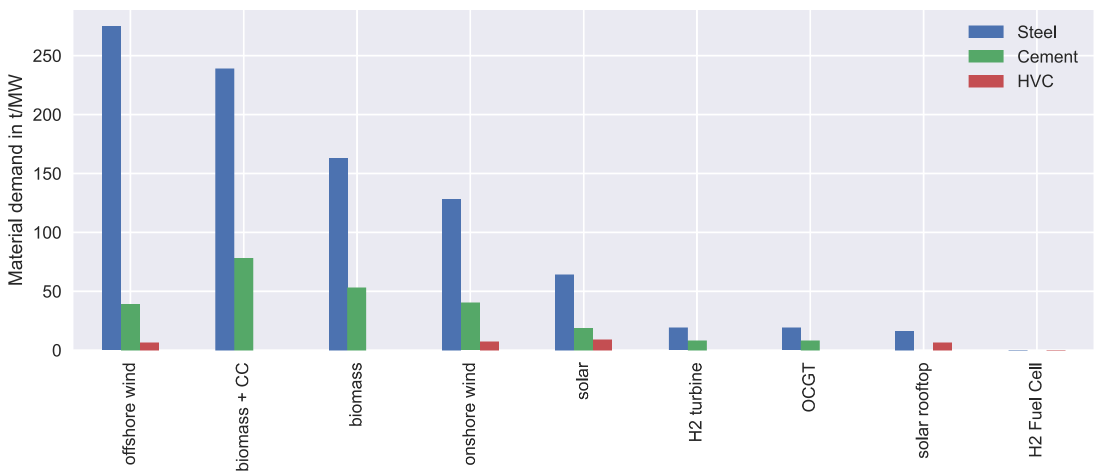

Fig. 3. Annualized material demand with equal distributed material demand over technology lifetime, for electricity generators, relative to capacity.

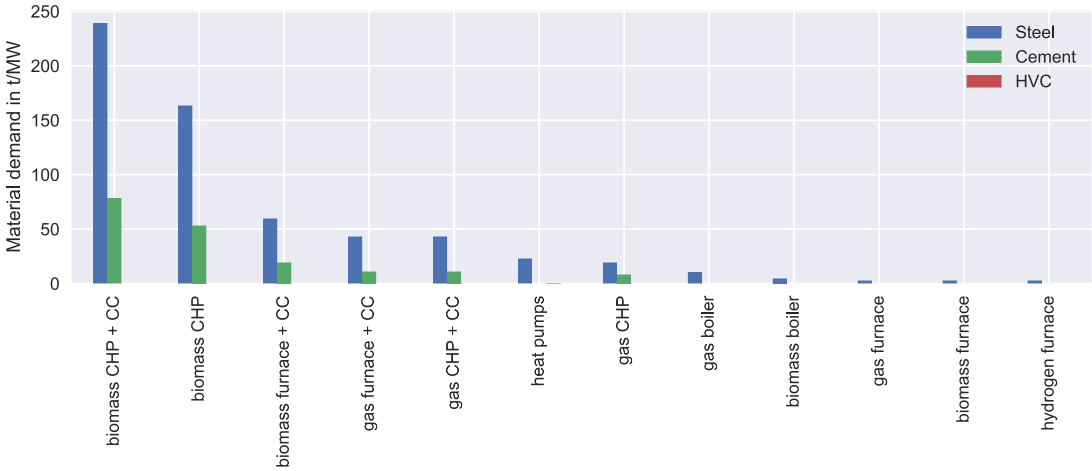

Fig. 4. Annualized material demand with equal distributed material demand over technology lifetime, for heat generators, relative to heat generation capacity.

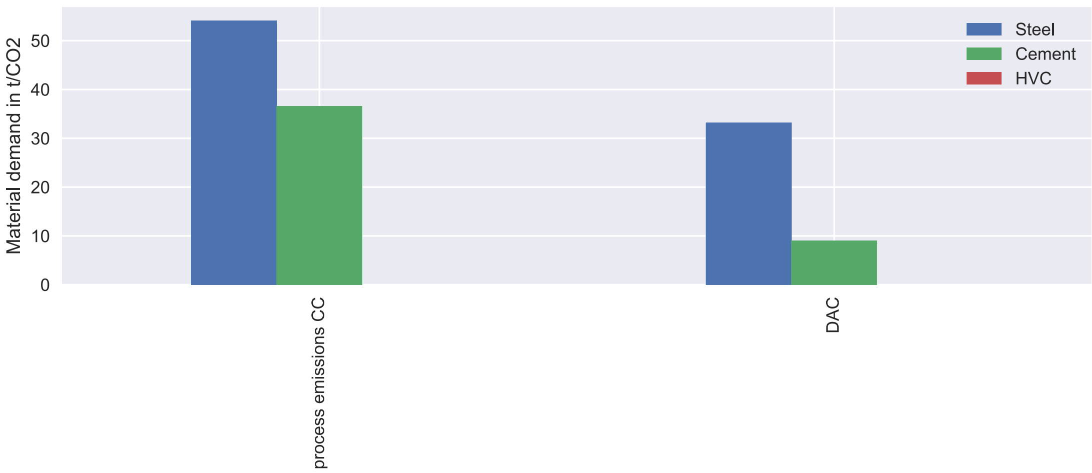

Fig. 5. Annualized material demand with equal distributed material demand over technology lifetime, for carbon capture technologies, relative to CO_2 capture capacity.

Appendix III
Results: H_2 use

fig.6 shows H_2 use and production at the German node for the reference scenario, the *100 % domestic* production scenario and the reference scenario with material demands from the *100 % domestic* production scenario. H_2 use increases compared to the reference scenario (307 TWh) in the two 100 % scenarios to similar levels (325 TWh *100 % domestic* and 327 TWh *increased-material reference*). Compared to the reference scenario, the *100 % domestic* scenario increases H_2 use for steel production (since more steel is produced) and additional Fischer-Tropsch (for HVC production) and additional H_2 turbines and fuel cells, while consumption by methanolisation and furnaces decrease (more biomass furnaces instead). In the increased-material reference scenario (100 %, not linked), compared to the reference scenario, technology choice remains equal with just more capacity expansion due to increased material demand, i.e. more H_2 for steel production (27 TWh compared to 20 TWh) and methanolisation (171 TWh compared to 166 TWh), but slightly less for H_2 furnaces (20 TWh compared to 22 TWh) due to a shift to biomass furnaces with BECCS. Also, more pipeline exports of H_2 to neighboring countries (16 TWh reference, 17 TWh *100 % domestic*, 25 TWh *increased-material reference*). Compared to the 100 % linked scenario, differences are that the production route for HVC remains methanol-to-olefines completely, such that only methanolisation and H_2 for steel use increases, and that no additional dispatchable electricity technology is expanded (fuel cell) and the H_2 for process heat is kept.

The increased H_2 demand compared to the reference is met by increased ship imports in the *100 % domestic* production scenario and with increased electrolysis in the *100 % domestic* production scenario. In the local

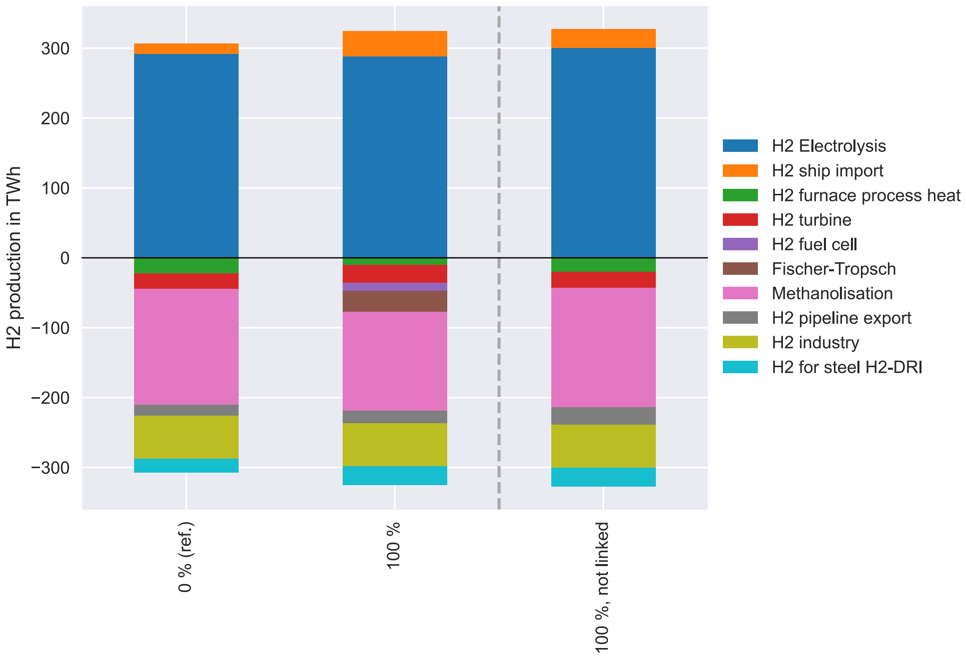

Fig. 6. Hydrogen production (positive values) and use (negative values) in the reference (0 %), *100 % domestic* (100 %), and increased-material reference (100 %, not linked) scenarios.

production scenario, electricity demand of electrolysis requires more expansion of generators requiring materials requiring industry expansion requiring generators expansion. In the unlinked scenario, the generators expansion does not have further effects. Ship imports of H_2 do not have an effect on capacity expansion or material demands since they come from outside the system.

Appendix IV
Detailed model description

Industry model: The industry model characterizes each process route and each process heat technology option with specific emissions, demand for energy carriers, process heat and raw materials (e.g., steel scrap or plastic waste for the recycling routes), and operating and annualised investment costs. The demand for process heat is partly endogenous because it depends on the expanded and operated process routes for bulk material production. Model inputs are bulk material demands (steel, cement, HVC) and process heat demand at different temperature levels for all industry branches.

Energy system model: The energy system model represents Germany and countries with a direct power transfer link, which is the same scope as used in the PyPSA-Ariadne version focusing on Germany [5], at a resolution of one node per country. Technologies for the conversion, transmission, and storage of multiple energy carriers (electricity, H_2, methane, naphtha and oil, methanol) and heat are expanded and operated. H_2 can be generated by electrolysis and exchanged between the countries via pipelines, or imported via ship. The exogenous industrial energy demand at the German node is replaced by attaching the industry model. The non-industrial energy demand, e.g., for buildings and transport, and the demand for industrial sectors other than steel, cement and HVC are exogenous inputs.

The operational costs of the industry processes do not include energy carrier costs, but energy carrier demands induce costs in the energy system to provide the energy carriers.

The capital costs of technologies include the costs for imported materials, while costs for locally produced materials are excluded, since they are indirectly represented by the necessary build-up and operation of industry processes (and energy system technologies). For the full-feedback case, the capital costs of technologies in PyPSA-Eur are corrected by subtracting material costs. The material costs are calculated based on material factors of a technology and commodity prices. These prices are taken from [6] and [7] and [8].

In the reference scenario, technology expansion at the German node and in neighboring countries is endogenous. In all other scenarios, technology capacities in neighboring countries are fixed to prevent shifts from material-demand-linked technologies in Germany to non-material-demand-linked technologies abroad. We do so to make the impact of the energy-to-material feedback identifiable.

TABLE II

VARIABLES

Variable	Index Sets	Description	Unit
$\overline{g}$	$\mathcal{K}$, $\mathcal{N}$	Installed capacity of generators	MW
$\overline{l}$	$\mathcal{E}$, $\mathcal{N}$, $\mathcal{N}$	Transmission line capacity node to node	MW
$\overline{i}$	$\mathcal{P}$, $\mathcal{N}$	Installed capacity of industrial processes	t/y
$\overline{f}$	$\mathcal{C}$, $\mathcal{N}$	Installed capacity of conversion technologies	MW
$\overline{b}^{ch}$	$\mathcal{S}$, $\mathcal{N}$	Installed charging capacity of storage	MW
$\overline{b}^{d}$	$\mathcal{S}$, $\mathcal{N}$	Installed discharging capacity of storage	MW
$\overline{e}$	$\mathcal{S}$, $\mathcal{N}$	Installed energy capacity of storage	MWh
b^{ch}	$\mathcal{S}$, $\mathcal{N}$, $\mathcal{T}$	charging of storage	MW
b^{d}	$\mathcal{S}$, $\mathcal{N}$, $\mathcal{T}$	discharging of storage	MW
e	$\mathcal{S}$, $\mathcal{N}$	energy of storage	MWh
g	$\mathcal{K}$, $\mathcal{E}$, $\mathcal{N}$, $\mathcal{T}$	Power production by generators	MWh
f	$\mathcal{C}$, $\mathcal{N}$, $\mathcal{T}$	Conversion flow	MWh
i	$\mathcal{P}$, $\mathcal{N}$, $\mathcal{T}$	Industry processes operation	t
l	$\mathcal{E}$, $\mathcal{N}$, $\mathcal{N}$, $\mathcal{T}$	Transmission of energy carrier from first node to second node	MWh
z^{ind}	-	Annual costs of industry	€
z^{esm}	-	Annual costs of energy system	€

objective function:

$$\min_{g,f,i,\overline{g},\overline{l},\overline{f},\overline{i},\overline{b^{ch}},\overline{b^{d}},\overline{e}} \quad z^{esm} + z^{ind} \tag{1}$$

with:

$$z^{ind} = \sum_{p\in\mathcal{P}}\sum_{n\in\mathcal{N}}\sum_{t\in\mathcal{T}} i_{p,n,t}\left(C_p^{\text{op,ind}} + \sum_{r\in\mathcal{R}} R_{p,r}^{\text{ind}} C_r^{\text{res}}\right) + \sum_{n\in\mathcal{N}}\sum_{p\in\mathcal{P}} \overline{i}_{p,n} C_p^{\text{inv,ind}} \tag{2}$$

$$\begin{aligned} z^{esm} = &\sum_{k\in\mathcal{K}}\sum_{n\in\mathcal{N}}\left(\sum_{t\in\mathcal{T}} g_{k,n,t} C_k^{\text{op,gen}} + \overline{g}_{k,e,n} C_k^{\text{inv,gen}}\right) \\ &+ \sum_{s\in\mathcal{S}}\sum_{n\in\mathcal{N}}\left(\overline{b}_{s,n}^{ch} C_s^{\text{inv,stor,ch}} + \overline{b}_{s,n}^{d} C_s^{\text{inv,stor,d}} + \overline{e}_{s,n} C_s^{\text{inv,stor,en}}\right) \\ &+ \sum_{e\in\mathcal{E}}\sum_{n,m\in\mathcal{N}} \overline{l}_{e,n,m} C_{e,n,m}^{\text{inv,tr}} \\ &+ \sum_{c\in\mathcal{C}}\sum_{n\in\mathcal{N}}\left(\sum_{t\in\mathcal{T}} f_{c,n,t}\left(C_c^{\text{op,conv}} + \sum_{r\in\mathcal{R}} R_{c,r}^{\text{conv}} C_r^{\text{res}}\right) + \overline{f}_{c,n} C_c^{\text{inv,conv}}\right) \end{aligned} \tag{3}$$

Energy balance:

$$D_{\text{e,n,t}} = \sum_{p\in\mathcal{P}} i_{\text{p,n,t}} \cdot E_{\text{p,e}} + \sum_{c\in\mathcal{C}} f_{\text{c,n,t}} \cdot E_{\text{c,e}} + \sum_{k\in\mathcal{K}} g_{\text{k,n,t}} \cdot E_{\text{k,e}} + \sum_{m\in\mathcal{N}} (l_{\text{e,m,n,t}} - l_{\text{e,n,m,t}}) + \sum_{s\in\mathcal{S}} (b_{\text{s,n,t}}^{\text{d}} - b_{\text{s,n,t}}^{\text{ch}}) \cdot E_{\text{s,e}} \quad \forall e\in\mathcal{E}, \forall n\in\mathcal{N}, \forall t\in\mathcal{T} \tag{4}$$

Material balance:

$$D_{\text{m,n}}^{\text{mat}} = \sum_{p\in\mathcal{P}}\sum_{t\in\mathcal{T}} i_{\text{p,n,t}} \cdot M_{\text{p,m}}^{\text{ind}} + \sum_{k\in\mathcal{K}} \overline{g}_{k,n} \cdot M_{\text{k,m}} + \sum_{s\in\mathcal{S}} \overline{e}_{s,n} \cdot M_{\text{s,m}} + \sum_{c\in\mathcal{C}} \overline{f}_{c,n} \cdot M_{\text{c,m}} \quad \forall n\in\mathcal{N}, \forall m\in\mathcal{M} \tag{5}$$

Resource constraint:

$$\sum_{p\in\mathcal{P}}\sum_{t\in\mathcal{T}} i_{p,n,t} \cdot R_{p,r} + \sum_{c\in\mathcal{C}}\sum_{t\in\mathcal{T}} f_{c,n,t} \cdot R_{c,r} \leq R_r^{max} \quad \forall r\in\mathcal{R}, \forall n\in\mathcal{N} \tag{6}$$

Emission constraint:

$$\sum_{p\in\mathcal{P}}\sum_{t\in\mathcal{T}} i_{p,n,t} \cdot X_p + \sum_{c\in\mathcal{C}}\sum_{t\in\mathcal{T}} f_{c,n,t} \cdot X_c + \sum_{k\in\mathcal{K}}\sum_{t\in\mathcal{T}} g_{k,n,t} \cdot X_k \leq 0 \quad \forall n\in\mathcal{N} \tag{7}$$

Operation constraints: The operation of all technologies is constrained by their capacities. The generation of renewable powerplants is further constrained by their time- and location-dependent capacity factors CF, which are defined between 0 and 1 as a fraction of capacity.

$$f_{c,n,t} \leq \overline{f}_{c,n} \quad \forall\, c\in\mathcal{C},\ \forall\, n\in\mathcal{N}, \forall\, t\in\mathcal{T} \tag{8}$$

TABLE III

SETS AND THEIR ELEMENTS. HIGHT = HIGH-TEMPERATURE, MEDIUMT = MEDIUM-TEMPERATURE, LOWT = LOW-TEMPERATURE.

Set	Description	Elements
$\mathcal{P}$	Industrial processes	'steel EAF', 'steel H_2-DRI+EAF', 'steel ISW', 'steel NG-DRI+EAF', 'hvc steam-cracker', 'hvc chemical recycling', 'hvc mechanical recycling', 'hvc electric steam-cracker', 'hvc MtO', 'cement CEM I', 'cement CEM II/C-M', 'cement CEM II/AB-M', 'cement CEM II C/ Q-L', 'lowT industry solid biomass', 'lowT industry solid biomass CC', 'lowT industry methane', 'lowT industry methane CC', 'lowT industry heat pump', 'lowT industry electricity', 'solid biomass for mediumT industry', 'solid biomass for mediumT industry CC', 'gas for mediumT industry', 'gas for mediumT industry CC', 'hydrogen for mediumT industry', 'gas for highT industry', 'gas for highT industry CC', 'hydrogen for highT industry', 'process emissions CC', 'gas for industry CC', 'mediumT industry electricity', 'plasma for highT industry', 'solid biomass for highT industry', 'highT industry solid biomass CC', 'coal for highT industry', 'waste for highT industry'
$\mathcal{M}$	Materials	Steel, cement, HVC
$\mathcal{S}$	Storage technologies	'CO_2 store', 'gas store', 'H_2 store', 'battery', 'residential rural water tanks', 'services rural water tanks', 'residential urban decentral water tanks', 'services urban decentral water tanks', 'urban central water tanks', 'methanol', 'oil', 'home battery', 'highT industry heat', 'mediumT industry heat', 'lowT industry heat'
$\mathcal{T}$	Timesteps	0, 3, ..., 2920 (1 year in 3h resolution)
$\mathcal{N}$	Nodes in the system	1 node per country for Germany + countries with direct power links
$\mathcal{C}$	Conversion technologies	'OCGT', 'H_2 Electrolysis', 'H_2 Fuel Cell', 'Sabatier', 'SMR CC', 'SMR', heat pump, resistive heater, gas boiler, gas CHP', gas CHP CC', 'biogas to gas', 'solid biomass CHP', ' solid biomass CHP CC', 'biomass boiler', 'biomass to liquid', 'BioSNG', 'methanolisation', 'Fischer-Tropsch', 'DAC', 'BioSNG CC', 'solid biomass to hydrogen CC', 'biomass to liquid CC', 'electrobiofuels', 'solid biomass to electricity', 'solid biomass to electricity CC', 'waste CHP CC', 'waste CHP', 'H_2 turbine'
$\mathcal{R}$	Resources	'biogas', 'solid biomass', 'municipal solid waste', 'plastic waste', 'packaging waste', 'steel scrap'
$\mathcal{K}$	Generation technologies	onshore wind, offshore wind, solar, solar rooftop, run-of-river, hydropower
$\mathcal{A}$	All technologies	$K \cup C \cup S \cup P$
$\mathcal{E}$	Energy carriers and heat	electricity, methane, hydrogen, space heat (residential / servives; central / decentral), industry process heat ('lowT industry', 'mediumT industry', 'highT industry'), methanol, oil / naphtha, coal
$\mathcal{E}^{\text{tr}}$	Transmittable energy carriers	Electricity, hydrogen

$$g_{k,n,t} \leq \overline{g}_{k,n} \cdot CF_{k,n,t} \quad \forall\ k \in \mathcal{K},\ \forall\ n \in \mathcal{N}, \forall\ t \in \mathcal{T} \tag{9}$$

$$l_{e,m,n,t} \leq \bar{l}_{e,m,n} \quad \forall\ e \in \mathcal{E}, \forall\ m \in \mathcal{N},\ \forall\ n \in \mathcal{N}, \forall\ t \in \mathcal{T} \tag{10}$$

Capacity limits:

$$0 \leq \overline{f}_{c,n} \leq \overline{F}_{c,n}^{max} \quad \forall\ c \in \mathcal{C},\ \forall\ n \in \mathcal{N} \tag{11}$$

$$\overline{G}_{k,n}^{start} \leq \overline{g}_{k,n} \leq \overline{G}_{k,n}^{max} \quad \forall\ k \in \mathcal{K},\ \forall\ n \in \mathcal{N} \tag{12}$$

$$0 \leq \bar{l}_{e,m,n} \leq \overline{L}_{e,m,n}^{max} \quad \forall\ e \in \mathcal{E},\ \forall\ n \in \mathcal{N}, \forall\ m \in \mathcal{M} \tag{13}$$

Transmission between nodes only possible for transmittable carriers:

$$\overline{L}_{e,m,n}^{max} = 0 \forall e \in \mathcal{E} \notin \mathcal{E}^{tr}, \forall m, n \in \mathcal{N} \tag{14}$$

We apply the optimal linear power flow method described in the PyPSA documentation [9]. Further constraints are described in the PyPSA documentation (*Storage Unit constraints, Generator constraints, Passive branch flows*). If not said otherwise, the default parameters of PyPSA-Eur are kept, e.g., investment costs, capacity factors, energy carrier limits.

Secondary material availability for HVC and steel in the year y = 2045 is calculated based on the method applied in [10] and [11]. Materials go to enduses, and become secondary material after the lifetime of the respective enduse $end \in Enduses$. The secondary material secondarymat in year y is $\text{secondarymat}_y = \sum_{end \in \text{Enduses}} (p_{(y-\text{lifetime}_{\text{end}})} \cdot \text{share}_{\text{end}} \cdot \text{recovery}_{\text{end}}$, with p being the production of the material in a specific year.

TABLE IV
PARAMETERS

Parameter	Index Sets	Description	Unit
$C^{op,ind}$	$\mathcal{P}$	Operating cost of industrial processes	€/t
$C^{inv,ind}$	$\mathcal{P}$	Investment cost of industrial processes	€/(t/y)
C^{res}	$\mathcal{R}$	Price of resources	€/MWh or €/t
R	$\mathcal{A}$, $\mathcal{R}$	Resource demand of technology	t/t or t/MWh
$C^{op,gen}$	$\mathcal{K}$	Operating cost of generators	€/MWh
$C^{inv,gen}$	$\mathcal{K}$	Investment cost of generators	€/MW
$C^{inv,stor,ch}$	$\mathcal{S}$	Investment cost for storage charging capacity	€/MW
$C^{inv,stor,d}$	$\mathcal{S}$	Investment cost for storage discharging capacity	€/MW
$C^{inv,stor,en}$	$\mathcal{S}$	Investment cost for storage energy capacity	€/MWh
$C^{inv,tr}$	$\mathcal{E},\mathcal{N},\mathcal{N}$	Investment cost for transmission	€/MW
$C^{op,conv}$	$\mathcal{C}$	Operating cost of conversion technologies	€/MWh
R_r^{max}	$\mathcal{R}$	Overall resource limit	t or MWh
$C^{inv,conv}$	$\mathcal{C}$	Investment cost of conversion technologies	€/MW
E	$\mathcal{A},\mathcal{E}$	Energy carrier production by technology (- when consumed)	MWh/MWh
M^{ind}	$\mathcal{P},\mathcal{M}$	Material production by technology	t/MWh
M	$\mathcal{A},\mathcal{M}$	Material consumption (- sign) by technology capacity	t/MW
D	$\mathcal{E},\mathcal{N},\mathcal{T}$	Exogenous energy carrier demand	MWh
$D^{non-ind}$	$\mathcal{E},\mathcal{N},\mathcal{T}$	Exogenous non-industrial energy carrier demand	MWh
D^{mat}	$\mathcal{M},\mathcal{N},\mathcal{T}$	Exogenous material demand	t
X	$\mathcal{A}$	Emissions from technology operation	t/MWh
X^{gen}	$\mathcal{K}$	Emissions of generators	t/MWh
$\overline{F}^{max}$	$\mathcal{C},\mathcal{N}$	Capacity limit of conversion technology c at n	MW
$\overline{G}^{max}$	$\mathcal{K},\mathcal{N}$	Capacity limit of generator k at n	MW
G^{start}	$\mathcal{K},\mathcal{N}$	Initial capacity of generator k at n	MW
$\overline{L}^{max}$	$\mathcal{E},\mathcal{N},\mathcal{N}$	Capacity limit of transmission from m to n	MW
CF	$\mathcal{K},\mathcal{N},\mathcal{T}$	Generator capacity factor	-

A. Input and model details

- Ship H_2 imports: We include H_2 import via ship transport to LNG terminals at a constant price of 84 €/MWh (taken from [8]). For the countries in scope, overseas pipeline imports are not an option [8].
- Flexibility of technologies: Industrial processes are modeled with a flat profile, i.e. are not flexible. Some conversion processes allow flexible operation, e.g., electrolysis (rampable 0% to 100% of capacity), methanolisation (50 % to 100 %), Fischer-Tropsch (90 % to 100%), process heat technologies (80 % to 100%), heat pumps and electric heaters and boilers (all 0 % to 100 %), dispatchable electricity generators (all 0 % to 100 %). No vehicle-to-grid is included.
- Exogenous material demand (not from energy technologies): We assume the same material production for Germany in 2045 as today (2021). Thus, effects of industry production moving to other countries or changes in material demand are not considered. This is changed in sensitivity analyses. Sources for material production today: steel and cement from the IDEES-2021 database [12]. The database gives only aggregated basic chemicals demand, so we take methanol and HVC production in Germany from [13].
- Spatial scope and resolution: The energy system model comprises one node per country for Germany and countries with a direct power transmission link (12 ”electricity neighbors” are represented, as in the model version of the Ariadne project [5]). Endogenous process route choice is only applied for the German node, while the simplified industry representation is kept for the other nodes. Process heat provision is endogenous for all nodes and industry branches (except for the German node in the soft-linked energy system optimisation where process heat provision is determined in the previous industry optimisation).
- Transmission: Transmission of energy carriers between nodes is only possible for electricity and hydrogen, through the expansion of lines, DC links and pipelines. CO_2, methane, oil / naphtha, and methanol used at the German node must also be provided by technologies at the German node, and resources cannot be imported from other nodes.
- Resource limits: Biomass, emissions and CO_2 storage capacity is limited for Germany. For the other nodes, joint limits are set. Biomass data for Germany and other countries in scope is taken from the JRC-ENSPRESO database, scenario ENS-Med, for the year 2040 [14]. Biomass to liquid and BioSNG are enabled as options. No additional (outside Europe) biomass imports are enabled.
- Temporal resolution and horizon: 3H resolution, 1 year.
- Exogenous energy demands (not from bulk material production): Electricity load for standard electricity applications and space heat (today’s level). Additional loads for electric vehicles are added. Electricity

demand for other power-to-X technologies is endogenous, e.g., heat pumps, electrolysers, methanolisation. Further exogenous energy demands: oil for aviation and agriculture, H_2 fuel cell, process heat, feedstock and electricity demand for other industry branches than steel, cement and HVC.

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